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In [27]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv("EDA sheet.csv")

# Q1: Maximum matches played by a player in a season
max_matches = df["Matches"].max() # Find the maximum matches played
players_with_max_matches = df[df["Matches"] == max_matches]["Player"].unique() # Get all players with max matches

print("Q1:" "Maximum matches played by a player in a season:")
for player in players_with_max_matches:
    print(player)
print(f"Total Matches: {max_matches}")

# Q2: Top 2 players with highest average who have at least 2 half-centuries
top_avg_players = df[df["50"] >= 2].nlargest(2, "Avg")[["Player", "Avg"]]
print("Q2:", top_avg_players)

# Q3: Create first and last name columns
df[["First Name", "Last Name"]] = df["Player"].str.split(n=1, expand=True)
print("Q3:", df[["First Name", "Last Name"]])

# Q4: Clean 'Highest Score' column
df['Cleaned_Highest_Score'] = df['Highest Score'].astype(str).str.replace('*', '', regex=False).astype(int)

# Check the cleaned column
print("Q4:", df[['Highest Score', 'Cleaned_Highest_Score']].head())

# Q5: Total number of centuries scored
total_centuries = df[df["100"] > 0]["100"].sum()
print("Q5:", f"Total Centuries: {total_centuries}")

# Q6: Players with strike rate less than average strike rate
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average_strike_rate = df["Strike rate"].mean()
low_sr_players = df[df["Strike rate"] < average_strike_rate][["Player", "Strike rate"]]
print("Q6:", low_sr_players)

# Q7: Correlation heatmap
plt.figure(figsize=(10, 6))

numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')

plt.title("Feature Correlation Heatmap")
plt.show()

# Q8: Players with average > 50 and strike rate > 120
high_performance_players = df[(df["Avg"] > 50) & (df["Strike rate"] > 120)]
print("Q8:", high_performance_players[["Player", "Avg", "Strike rate"]])

# Q9: Players with avg > 40 and balls faced > 100
consistent_players = df[(df["Avg"] > 40) & (df["Balls faced"] > 100)]
print("Q9:", consistent_players[["Player", "Avg", "Balls faced"]])

# Q10: Players with at least one century
century_players = df[df["100"] >= 1][["Player"]]
plt.figure(figsize=(8, 4))
sns.countplot(y=century_players, order=century_players.value_counts().index)
plt.title("Players with at least one century")
plt.show()

# Q11: Players with at least 4 half-centuries
half_century_players = df[df["50"] >= 4][["Player"]]
print("Q11:", half_century_players)

# Q12: Players with more than 45 boundaries and 10 sixes
boundary_six_players = df[(df["4s"] > 45) & (df["6s"] > 10)][["Player", "4s", "6s"]]
print("Q12:", boundary_six_players)

# Q13: Histogram of matches played
num_bins = 10 # Number of bins
counts, bins, patches = plt.hist(df['Matches'], bins=num_bins, edgecolor='black')
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colors = plt.cm.viridis(np.linspace(0, 1, num_bins))
for patch, color in zip(patches, colors):
    patch.set_facecolor(color)

plt.xlabel('Matches Played')
plt.ylabel('Frequency')
plt.title('Histogram of Matches Played')
plt.show()

# Q14: Histogram of balls faced
plt.hist(df["Balls faced"], bins=10, color='blue', edgecolor='black')
plt.xlabel("Balls faced")
plt.ylabel("Frequency")
plt.title("Distribution of Balls faced by Players")
plt.show()

# Q15: Top 10 players with most runs
top_run_scorers = df.nlargest(10, "Runs")[["Player", "Runs"]]
print("Q15:", top_run_scorers)

# Q16: Players who played but didn't bat
non_batting_players = df[df["Inns"] == 0][["Player"]]
print("Q16:", non_batting_players)

# Q17: Percentage of total runs from 4s and 6s
df["Runs_from_Boundaries"] = df["4s"]*4 + df["6s"]*6
df["Boundary_Percentage"] = (df["Runs_from_Boundaries"] / df["Runs"]) * 100
top_boundary_players = df.nlargest(5, "Boundary_Percentage")[["Player", "Boundary_Percentage"]]
print("Q17:", top_boundary_players)

# Q18: Top 5 players with highest Not Out %
df["Not_Out_Percentage"] = (df["Not Out"] / df["Inns"]) * 100
top_not_out_players = df.nlargest(5, "Not_Out_Percentage")[["Player", "Not_Out_Percentage"]]
print("Q18:", top_not_out_players)

# Q19: Top 10 players with most sixes
top_six_hitters = df.nlargest(10, '6s')[['Player', '6s']]
plt.figure(figsize=(10,5))
sns.barplot(x='Player', y='6s', data=top_six_hitters)
plt.xticks(rotation=90)
plt.title("Top 10 Players with Most Sixes")
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plt.show()

# Q20: Scatter plot of runs vs balls faced
plt.scatter(df['Balls faced'], df['Runs'], color='purple', alpha=0.6)
plt.xlabel('Balls Faced')
plt.ylabel('Runs Scored')
plt.title('Runs vs Balls Faced Scatter Plot')
plt.show()

# Checking correlation between Runs and Balls faced
correlation = df[['Balls faced', 'Runs']].corr()
print("Q20: Correlation between Runs & Balls Faced:\n", correlation)
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Q1:Maximum matches played by a player in a season:

Shikhar Dhawan

Shreyas Iyer

Marcus Stoinis

Kagiso Rabada

Total Matches: 17

Q2: Player Avg

36 Wriddhiman Saha 71.33

4 Ishan Kishan 57.33

Q3: First Name Last Name

0 KL Rahul

1 Shikhar Dhawan

2 David Warner

3 Shreyas Iyer

4 Ishan Kishan

..

128 Khaleel Ahmed

129 Arshdeep Singh

130 Daniel Sams

131 Shreevats Goswami

132 Trent Boult

[133 rows x 2 columns]

Q4: Highest Score Cleaned_Highest_Score

0 132* 132

1 106* 106

2 85* 85

3 88* 88

4 99 99

Q5: Total Centuries: 5

Q6: Player Strike rate

51 Ajinkya Rahane 105.60

55 Glenn Maxwell 101.88

58 Vijay Shankar 101.04

61 Josh Philippe 101.29

62 Gurkeerat Singh 88.75

65 Kedar Jadhav 93.93

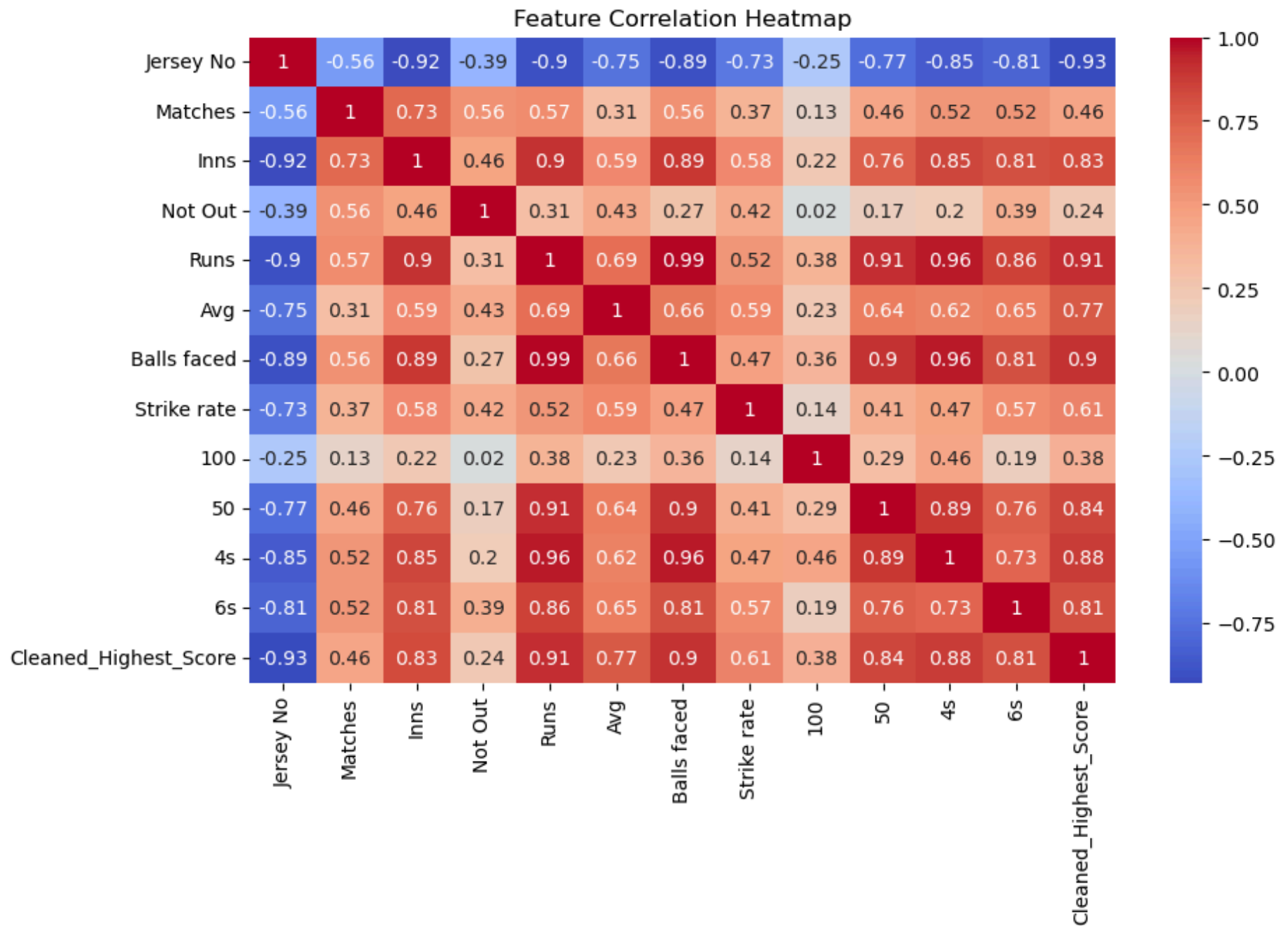
70 Yashasvi Jaiswal 90.90

71 Shreyas Gopal 94.87

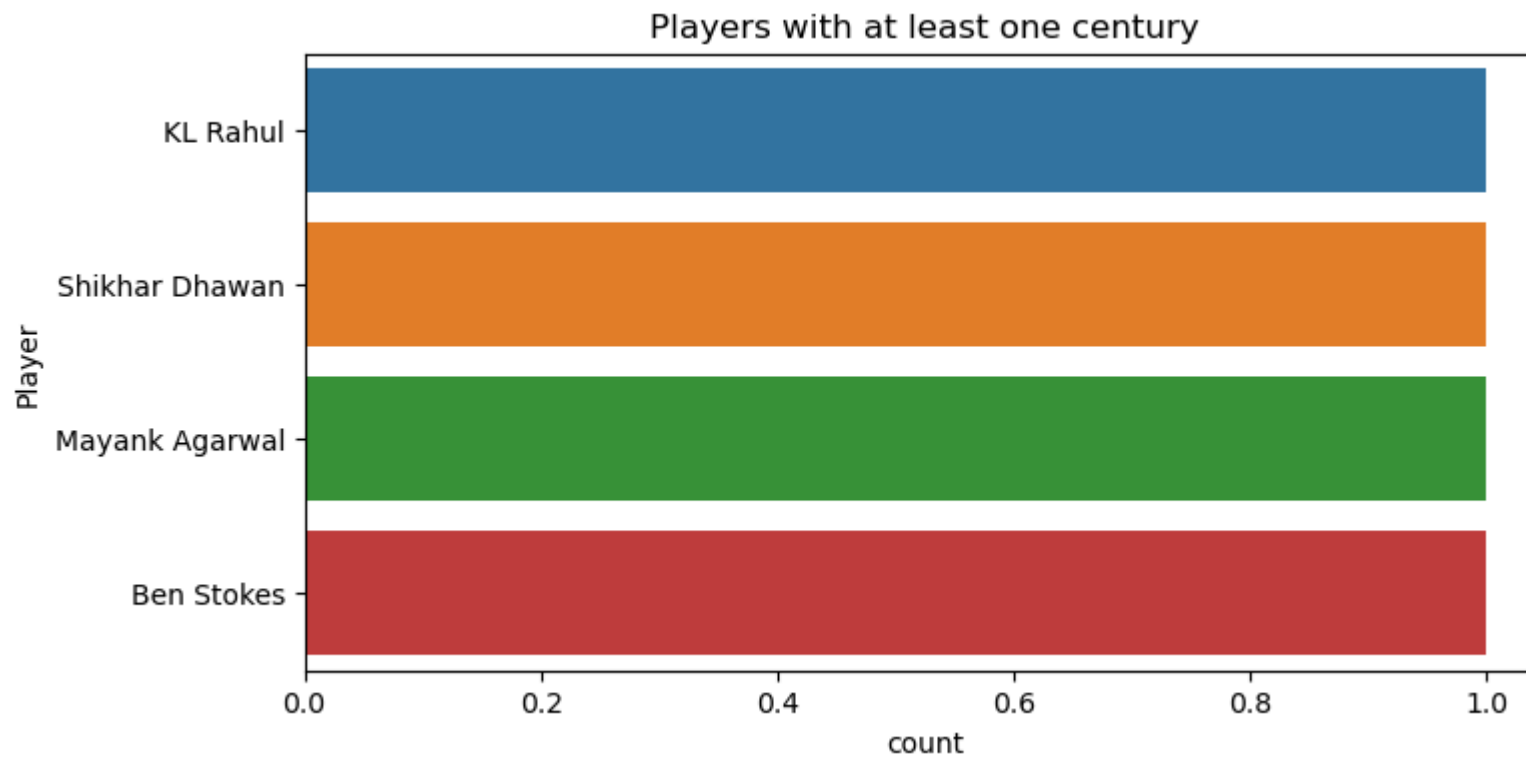
77 Murali Vijay 74.41

79 Chris Jordan 93.54

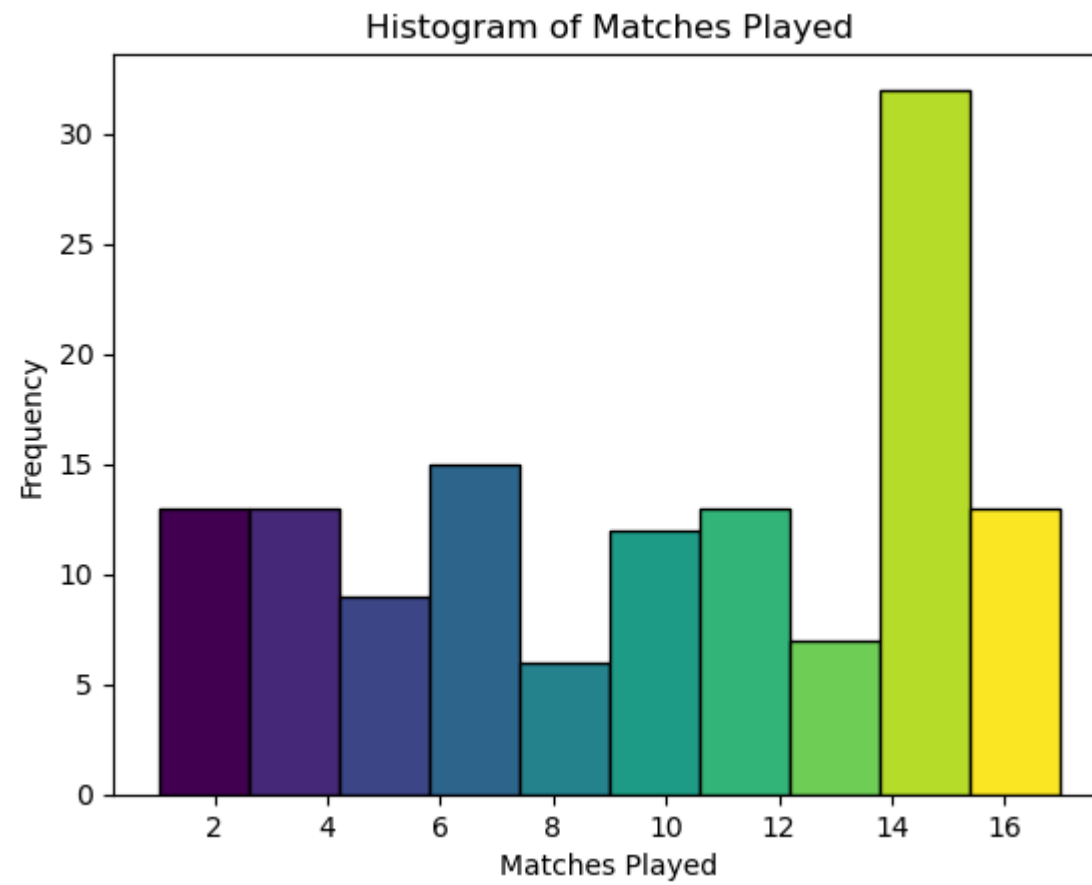
80	Navdeep Saini	100.00
82	Kamlesh Nagarkoti	70.96
84	Harshal Patel	87.50
85	Jimmy Neesham	105.55
86	Tom Banton	90.00
89	Prabhsimran Singh	100.00
92	Kuldeep Yadav	61.90
94	Moeen Ali	75.00
95	Sandeep Sharma	80.00
96	Shardul Thakur	57.14
98	Rinku Singh	100.00
99	Shivam Mavi	71.42
100	Varun Chakaravarthy	66.66
101	Jaydev Unadkat	69.23
102	Ankit Rajpoot	90.00
104	Shahbaz Nadeem	87.50
105	Pravin Dubey	53.84
107	Deepak Chahar	58.33
108	Ravi Bishnoi	58.33
109	Andrew Tye	100.00
111	Kartik Tyagi	66.66
112	Murugan Ashwin	100.00
114	T Natarajan	60.00
115	Prasidh Krishna	50.00
116	Rahul Chahar	50.00
117	Mohammad Shami	66.66
118	Nikhil Naik	33.33
119	Mujeeb Ur Rahman	33.33
120	Dale Steyn	50.00
121	Varun Aaron	10.00
122	Shahbaz Ahmed	100.00
123	Yuzvendra Chahal	33.33
124	Mitchell Marsh	0.00
125	Umesh Yadav	0.00
126	Bhuvneshwar Kumar	0.00
127	Sheldon Cottrell	0.00
128	Khaleel Ahmed	0.00
129	Arshdeep Singh	0.00
130	Daniel Sams	0.00
131	Shreevats Goswami	0.00
132	Trent Boult	0.00

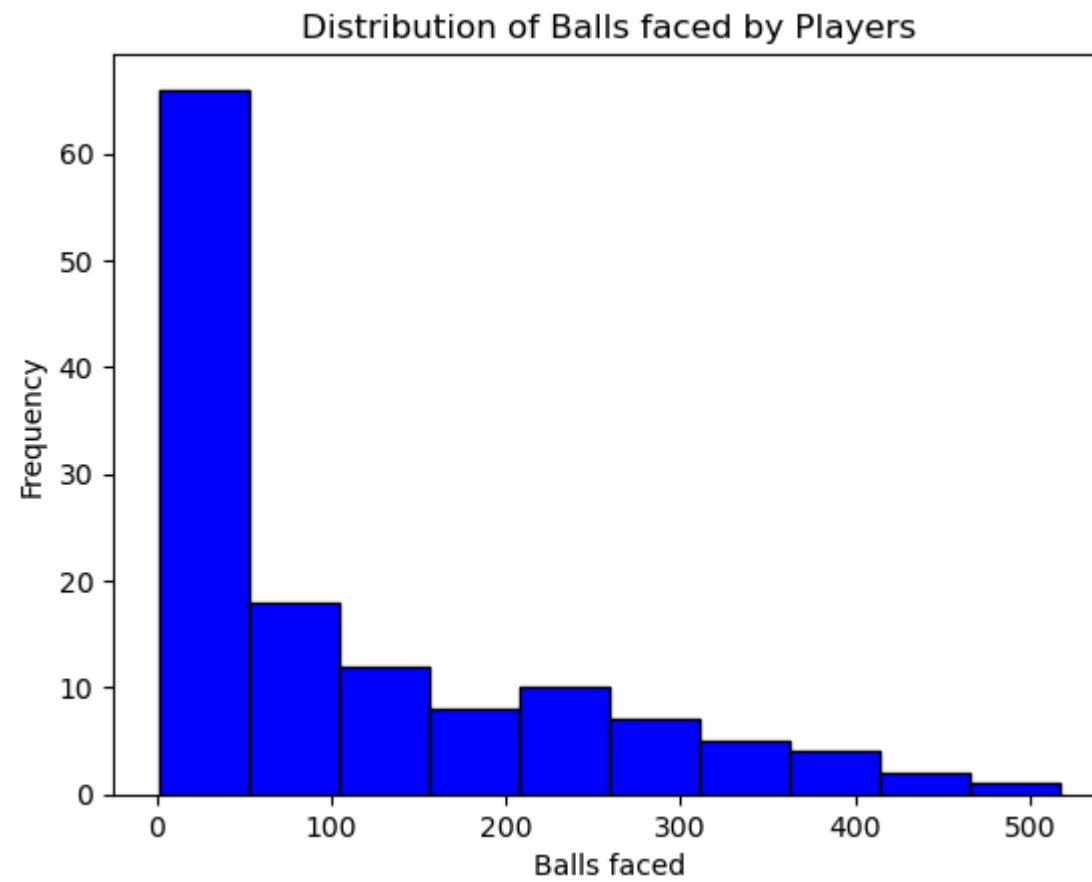


Q8:	Player	Avg	Strike rate
0	KL Rahul	55.83	129.34
4	Ishan Kishan	57.33	145.76
31	Kieron Pollard	53.60	191.42
36	Wriddhiman Saha	71.33	139.86
37	Ruturaj Gaikwad	51.00	120.71
57	Deepak Hooda	101.00	142.25
60	Tom Curran	83.00	133.87
Q9:	Player	Avg	Balls faced
0	KL Rahul	55.83	518
1	Shikhar Dhawan	44.14	427
4	Ishan Kishan	57.33	354
8	Virat Kohli	42.36	384
9	ABD Villiers	45.40	286
10	Faf Duplessis	40.81	319
14	Eoin Morgan	41.80	302
24	Kane Williamson	45.28	237
27	Chris Gayle	41.14	210
28	Ben Stokes	40.71	200
31	Kieron Pollard	53.60	140
32	Rahul Tewatia	42.50	183
33	Ravindra Jadeja	46.40	135
36	Wriddhiman Saha	71.33	153
37	Ruturaj Gaikwad	51.00	169

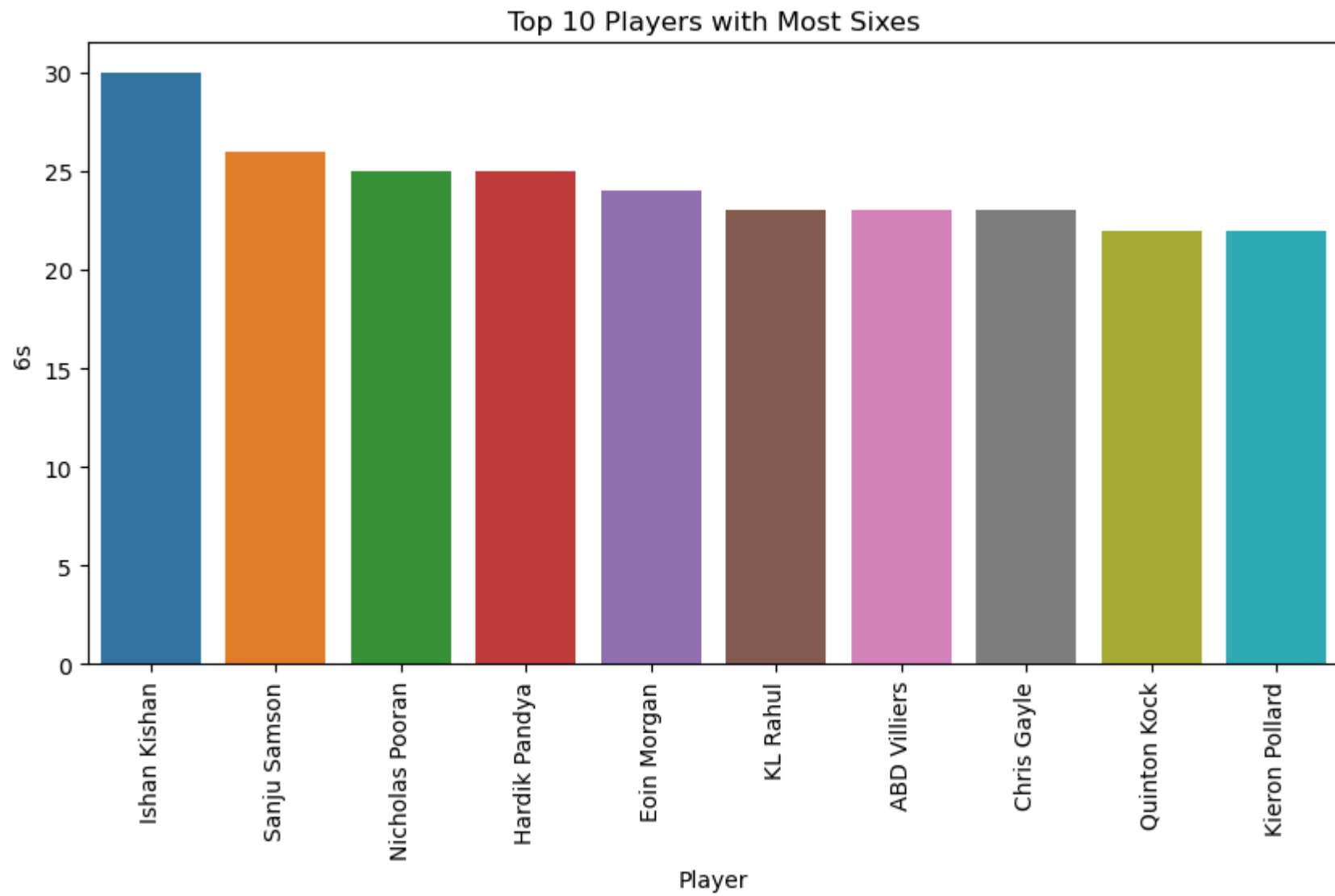


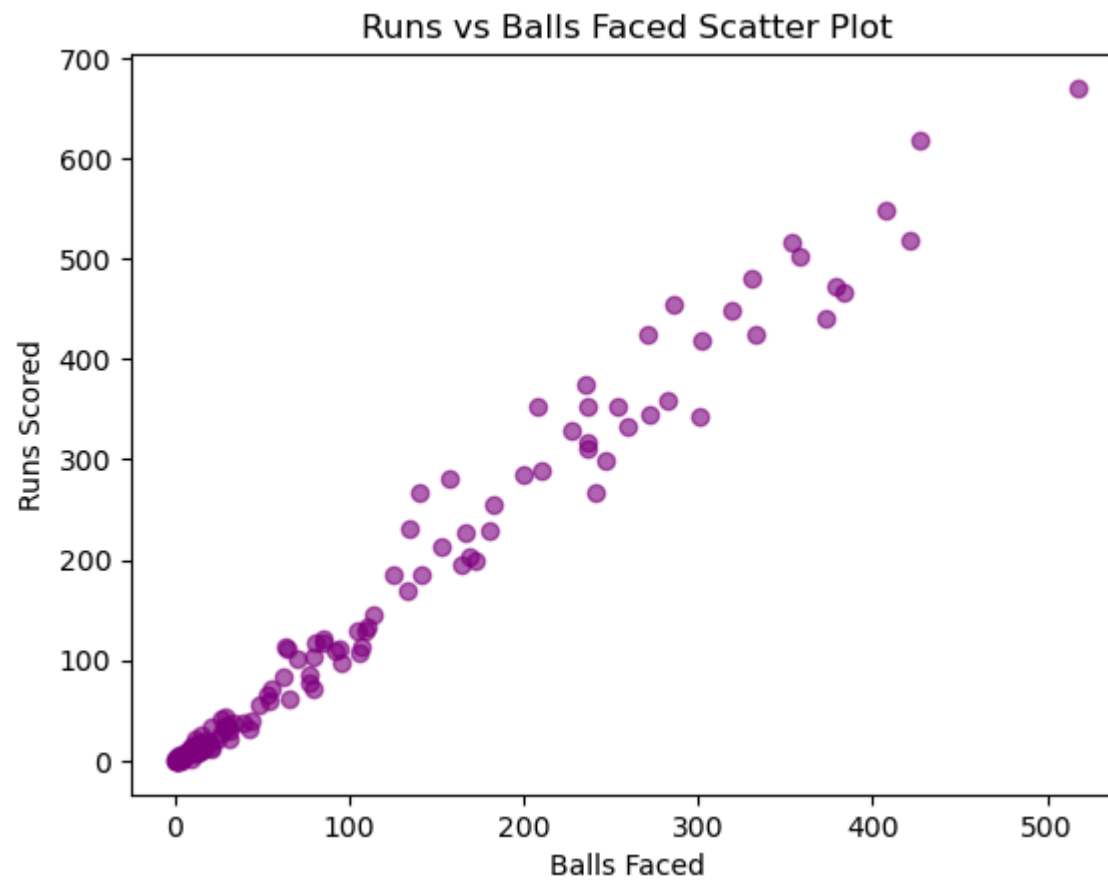
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Q11: 0          KL Rahul
1      Shikhar Dhawan
2      David Warner
4      Ishan Kishan
5      Quinton Kock
6      Suryakumar Yadav
7      Devdutt Padikkal
9      ABD Villiers
10     Faf Duplessis
Name: Player, dtype: object
Q12:          Player  4s  6s
0          KL Rahul  58  23
1      Shikhar Dhawan  67  12
2      David Warner  52  14
5      Quinton Kock  46  22
6      Suryakumar Yadav  61  11
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Q15:          Player  Runs
0      KL Rahul    670
1    Shikhar Dhawan    618
2      David Warner    548
3      Shreyas Iyer    519
4      Ishan Kishan    516
5      Quinton Kock    503
6    Suryakumar Yadav    480
7    Devdutt Padikkal    473
8      Virat Kohli    466
9      ABD Villiers    454
Q16: Series([], Name: Player, dtype: object)
Q17:          Player  Boundary_Percentage
109    Andrew Tye          100.000000
48    Andre Russell          76.923077
74    Chris Morris          76.470588
29    Hardik Pandya          73.309609
47    Sunil Narine          72.727273
Q18:          Player  Not_Out_Percentage
62    Gurkeerat Singh          100.0
68    Lockie Ferguson          100.0
93      Imran Tahir          100.0
97    Mohammad Nabi          100.0
105    Pravin Dubey          100.0
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Q20: Correlation between Runs & Balls Faced:

	Balls faced	Runs
Balls faced	1.000000	0.989948
Runs	0.989948	1.000000

In []: